Energy Star Home Performance Contracting Program Takes Off


NYSERDA

Energy Star has already made its name through programs for promoting consumer purchases of energy-efficient products and for building energy-efficient new homes. Now home performance contractors in New York can also partner up to benefit from this great initiative.

by Andrew Fisk

The Hickok family of Weedsport, New York, was one of the first customers to have a comprehensive package of building performance work done on their home by a certified contractor through a new state initiative, Home Performance with Energy Star. Their home was 39 years old, used a lot of energy, and was uncomfortable to live in. After a thorough assessment that included a blower door test and a comprehensive combustion safety evaluation, the Hickoks invested in a package of home improvements that will enhance the comfort and durability of their home while saving them money on their energy bills.

Rx for Poorly Performing Homes

Many homes, like the Hickoks', suffer from unnecessary health, safety, and comfort problems due to improperly installed mechanical systems, inadequate insulation levels, and uncontrolled air infiltration (see "Sick Houses: Using Diagnostic Tools to Improve Indoor Air Quality," *HE* Nov/Dec '97, p. 15). However, in the past it has been difficult to convince consumers and residential contractors that these problems are diagnosable and treatable. Commonly held assumptions—for instance, that all homes have drafts, that a little mold is normal, or that uncomfortable humidity swings from season to season are inevitable—need to be dispelled and replaced with the assertion that people do not have to live with these problems. Home Performance with Energy Star now provides contractors with the knowledge and tools to recognize and help to resolve common house problems such as ice dams, temperature swings, mold and mildew, poor indoor air quality, peeling paint, and structural damage.

Under the Home Performance with Energy Star initiative, which was launched in February under the auspices of the New York Energy Smart program and sponsored by the New York State Energy Research and Development Authority (NYSERDA), contractors are acquiring the knowledge, skills, and equipment needed to diagnose such problems. They will be able to improve the delivery of services to their customers to the point where efficiency, comfort, and safety are the norm. This initiative aims to promote consumer awareness and demand for whole-house testing and treatments for energy efficiency, and to build a network of certified home performance contractors who are equipped to diagnose and improve energy efficiency and overall home performance.

Top-to-Bottom Improvements

One of the Hickoks' home improvements was reduced air leakage through the building shell thanks to added cellulose insulation in the walls and attic, installation of new Energy Star–labeled windows, and air sealing throughout the home. The air sealing measures included application of caulk and foam products, weatherstripping, gaskets, some Tyvek wrap, and dense-packed cellulose insulation. These measures had the effect of sealing off a potentially dangerous air leakage path between the attached garage and the living space via the ceiling cavity. High-density cellulose insulation was installed in strategic areas to reduce the air flow between interstitial spaces, including the cantilevered roof of a breezeway. The attic received blown-cellulose insulation to a value of R-44. Postretrofit blower door tests revealed that the insulation and air sealing measures reduced air infiltration by more than 40%.

Once the shell improvements were completed, a 90% AFUE sealed-combustion furnace was installed to replace the existing 39-year-old system. A heat load calculation determined that the improvements to the shell allowed the new furnace to be 20,000 Btu per hour input smaller than the old system and still provide ample heating. The existing water heater was replaced with a high-efficiency 40-gallon Bradford White gas water heater, and an inadequate pitch on the water heater flue pipe was corrected.

Since the Hickoks had previously decided to install a new central cooling system in their home, they were now motivated to ensure that the new cooling system would be as efficient as the rest of the house. Based on this decision, the family purchased an Energy Star-labeled 12-SEER cooling system instead of the 10-SEER system they had planned to install.

Along with major upgrades to the home's envelope, HVAC system, and appliances, the Hickoks also replaced the entire roof and siding to protect their investment in energy-related shell improvements. By using the low-interest Energy Star financing option available through the initiative, the Hickoks paid only 5% interest on their $20,000 loan.


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Steve Cowell of CSG explains how to use an infrared camera to look at insulation levels within the walls of a house.

Contractor Opportunities

To date, more than 50 homes like the Hickoks' have been improved through NYSERDA's initiative. But these families are not the only beneficiaries of this new Energy Star initiative. The contractors who take part harvest a whole crop of benefits, from training to new referrals. This is an important part of the Home Performance with Energy Star initiative—the development of an infrastructure of contractors who are educated, trained, and certified in the fundamentals of building science, building performance testing, and whole-house principles and applications. With training, certification, and the necessary diagnostic tools, contractors can differentiate themselves from the competition, marketing and selling home performance testing and treatments to homeowners in a comprehensive way. (For more about home performance training and certification, see "Defining a Whole-House Contractor," *HE* Jul/Aug '01, p. 22, and "Training for Tomorrow: Are Your Contractors Certifiable?" *HE* Jan/Feb '99, p. 29.)

Home performance contractor training is provided in New York by several organizations, including Conservation Services Group (CSG), the Select Board of Cooperative Educational Services (BOCES), and the Building Performance Contractors Association (BPCA). Once trained, contractors must get certified before they can participate in the Home Performance with Energy Star initiative.

The Building Performance Institute (BPI) is the national certification and standard-setting organization for the Home Performance with Energy Star initiative. BPI, in cooperation with building science experts across the country, defined the integrated testing procedures and certification standards for the contractors participating in the Home Performance with Energy Star initiative. Participating contractors elect to be certified as auditors, building shell specialists, and/or heating specialists, depending on their typical job duties and the corporate structure of their company. BPI coordinates with the training organizations to ensure that the training programs prepare contractors for BPI certification tests. It takes four to six weeks to complete the training and certification process.

To date, more than 30 home improvement firms are participating in the New York initiative, including some insulation and heating companies and a few established building performance auditing and contracting companies. More than 60 technicians have been certified by BPI, and the number is growing daily. Program managers had set an initial goal of 110 accredited firms in the state by the end of this year. Based on early results, they anticipate that this goal will be achieved. Most of the home performance contractor trainees at the organizations listed above are working toward participation in the NYSERDA Energy Star initiative, and it looks as if their efforts will quickly pay off. Consumer demand for the initiative is so high that contractors already in the program have more leads than they can follow up on; they are passing leads on to new trainees who can do the work right away.

Many firms and technicians have completed certification in both heating and shell specialization to enable them to work as general contractors and oversee subcontracting companies for the installation of energy efficiency measures. Perhaps more importantly, insulation and heating contractors are partnering on projects to provide comprehensive whole-house services to help increase business and customer satisfaction, while continuing to work within their individual areas of expertise. This cooperative effort between contractors not only ensures that the house-as-a-system concept is addressed, but it also increases profitability for the firms, as customer leads are shared.

The Certification Process

BPI certification requires successful completion of both a written, knowledge-based exam and a field performance-based exam. This ensures that technicians demonstrate a mastery of the specific skills associated with the application of building science principles to traditional building trade practice. While most certifications in other industries require only a written exam, BPI believes that the ability to perform tasks and to understand how best to apply test results is key to a technician's ability to inspect and diagnose a home.

The field evaluation requires that candidates demonstrate proficiency in inspecting, diagnosing, and specifying improvements to homes in a real-world environment. The field exams are held in an actual home, where the candidate must perform all the functions required for the certification in a proctored environment.

Home performance contractors trained for the New York initiative can work toward two levels of certification—Technician I (Tech I), and Technician II (Tech II). To participate in the New York Home Performance with Energy Star initiative, contractor companies must have staff certified at the Tech II level. This level is broken down into Tech II Shell Specialist and Tech II Heating Specialist. The specialists are only allowed to oversee work that falls under their specialty area.

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Mark Hutchins of CSG shows how to do a carbon monoxide leakage test.

Technician I: Auditor

This is the first step in becoming a BPI certified specialist. Many firms use this level of certification for their sales staff. The requirements for Tech I include the tools needed to perform a basic yet comprehensive audit on an existing home:

- fundamentals of building science;
- whole-house auditing inspection skills;
- combustion safety testing skills;
- basic blower door testing skills; and
- fundamentals of health and safety.

Technician II: Shell Specialist

This certification is recommended for the crew chief or manager of an insulation or remodeling company. To be certified as a Tech II: Shell Specialist, candidates must first achieve Tech I certification, then also demonstrate their knowledge of the following subjects:

- moisture management strategies;
- ducted distribution systems;
- advanced blower door applications;
- specifications for mechanical ventilation;
- thermal imaging techniques;
- definition of the building envelope;
- air sealing specifications and installation;
- insulation specifications and installation;
- high-density insulation strategies and techniques;
- window and door specifications and installation;
- postinstallation diagnostics and verification; and
- additional health and safety considerations.

Technician II: Heating Specialist

This certification is recommended for the crew chief or manager of an HVAC or heating and plumbing business. To be certified as a Tech II: Heating Specialist, candidates must first achieve Tech I certification, then also demonstrate their knowledge of the following subjects:

- the science of heating systems;
- distribution system design;
- heating system sizing;
- combustion safety inspection and mitigation;
- heating system controls;
- distribution system inspections;
- domestic hot water system efficiency and safety evaluation;
- heating system efficiency tests;
- furnace heat exchanger inspections;
- ducted distribution system diagnostic tests for air flow, leakage, and pressure balancing;
- heating appliance cleaning and tune-up;
- combustion appliance venting;
- duct sealing;
- postinstallation diagnostics and

verification; and
- additional health and safety concerns.

The Experience in New York

NYSERDA, whose primary mission is to help lower energy costs for New Yorkers, has been working with a network of energy efficiency experts, contractors, and building scientists over the last three years to develop a strategy and implementation plan to improve the market for building performance contracting. The initiative is based on introducing home improvement contractors to the concept of looking at the house as a system, and the value of home performance diagnostic testing procedures. The challenge for NYSERDA is to strike a balance among training, financial incentives, and marketing to create a viable business model for participating contractors.

Through subsidies of up to 75% for training, certification costs, and equipment packages, NYSERDA has reduced the out-of-pocket expenses for contractors entering the building performance business from an estimated $12,000–$15,000 down to $2,000–$3,000.

It also offers 10% down, 0% interest agreements for the purchase of diagnostic equipment, including laptop computers, blower doors, duct testing equipment, CO meters, combustion gas analyzers, flow hoods, and HVAC system design software (WrightSoft *Manual J*).

NYSERDA will also provide computerized home assessment software with free training and technical support. It offers contractors financial incentives for installing energy-saving measures and appliances, for referring jobs after performing a whole-house audit, and for multitrade installations.

Contractor Marketing

In addition to the contractor incentives outlined above, NYSERDA has worked with CSG to develop a major marketing campaign to help contractors lower their costs for generating a consumer base. These efforts include:
- a $2 million-plus multimedia advertising and public relations campaign for Home Performance with Energy Star, featuring Steve Thomas, the host of This Old House, with TV, radio, newspaper, direct mail, and Web exposure;
- special events, including home shows, outreach and presentations to local trade organizations, and a regional Affordable Comfort conference held in Syracuse; and
- co-op advertising funding for participating contractors.

Happy Customers

Ultimately, the best way of promoting this new Energy Star initiative will be through word of mouth, as more and more families are able to fight rising energy costs through home performance improvements—protecting their home investment and ensuring home safety at the same time. The Hickoks, for example, can expect to save up to 50% in heating and cooling costs on their improved home. "The savings from the home improvements couldn't come at a better time," says Hickok, "with our first baby on the way."

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